

Electron Predictions in the Field–Wave Approach (FWA): Why Dynamic Algebra Greatly Exceeds Linear Models

In standard quantum theory, the electron is treated as a point particle with fixed parameters: charge, mass, spin, and Compton wavelengths. These quantities are inserted as external constants. In contrast, the Field–Wave Approach (FWA) describes the electron as a stable field node, whose properties arise from:

- the distinguishability depth L_u ,
- the Planck background,
- the local structure of the field.

This dynamic interpretation allows FWA to derive electron parameters rather than assume them. As a result, FWA achieves orders of magnitude higher accuracy than linear algebraic models.

1. Central FWA Relation

The fundamental scaling relation in FWA is:

$$\lambda_e = \lambda_P / \sqrt{L_u}$$

Unlike standard quantum theory, FWA uses the full Compton wavelength:

$$\lambda_C = h / (m_e c),$$

instead of the reduced Compton wavelength:

$$\bar{\lambda}_C = \hbar / (m_e c).$$

With:

- $\lambda_P = 1.616255 \times 10^{-35} \text{ m}$,
- $L_u \approx 4.44 \times 10^{-47}$,

FWA predicts:

$$\lambda_e^{\text{FWA}} \approx 2.426310 \times 10^{-12} \text{ m},$$

which matches the measured Compton wavelength to about ten significant digits.

This agreement is not used as a definition. FWA requires that L_u be derived independently from node stability on the Planck background. Thus:

$$L_u = (\lambda_P / \lambda_C)^2$$

is a consistency check, not a defining equation.

2. Derived Electron Parameters

Once λ_e is known, all other electron parameters follow directly:

$$m_e = h / (\lambda_e c), \quad k_e = 2\pi / \lambda_e, \quad \omega_e = 2\pi c / \lambda_e.$$

Thus, in FWA, the electron mass is not a fixed constant, but a field-dependent quantity emerging from the node structure.

3. Locality of Lu and Variable Electron Properties

A key prediction of FWA is that Lu becomes a local field quantity:

$$Lu = Lu(r, t, \Phi, \nabla\Phi, \dots)$$

Therefore:

$$\lambda_e(r, t) = \lambda_p / \sqrt{Lu(r, t)}, \quad m_e(r, t) = h\sqrt{Lu(r, t)} / (c \lambda_p).$$

This means:

- the effective electron mass,
- the electron wavelength,
- and the interaction strength

can change depending on the field configuration.

Standard quantum theory does not allow such variations — electron parameters are strictly invariant.

4. Electron–Electron Interaction in FWA

In FWA, electron interaction is not defined by an abstract charge number. Instead, it arises from the dynamics of field nodes.

The external momentum flux:

$$J = -\kappa_c W \nabla W$$

naturally produces repulsion between nodes.

Changes in Lu or field structure modify the node shape and therefore the interaction. In complex topological fields, the charge-flow contour may:

- close → producing screening,
- stretch → producing enhancement.

The corresponding topological flux can be expressed as a loop integral of the field potential, whose normalization defines the effective charge.

5. Anomalous Magnetic Moment

For the anomalous magnetic moment:

$$a_e = (g - 2) / 2,$$

FWA interprets $g - 2$ as a consequence of node field modes.

A complete FWA theory must reproduce the QED expansion:

$$a_e = (\alpha / 2\pi) + C_2(\alpha / \pi)^2 + C_3(\alpha / \pi)^3 + \dots$$

and identify FWA modes corresponding to these coefficients. This provides a direct mathematical test against precision QED.

6. Experimental Predictions

Because Lu is local, FWA predicts measurable changes in:

- electron wavelength,
- electron mass,
- interaction strength,

in environments such as:

- plasmas,
- atoms,
- superconductors,
- extreme electromagnetic fields.

These effects distinguish FWA from models where electron parameters remain strictly invariant.

7. Why FWA Is Far More Accurate Than Linear Algebra

Linear algebraic models are static:

$$u' = M u.$$

They cannot derive:

- electron mass,
- electron wavelength,
- interaction strength,

- anomalous magnetic moment.

They can only assign these values.

FWA is dynamic:

$$\partial_t u = -Au + \Omega(u) + \Phi(u),$$

and derives all electron properties from field structure.

The accuracy difference is enormous:

- Linear scaling from Planck to electron scale yields errors of $10^{20} - 10^{25}$.
- FWA matches experiment to 10^{-10} .

Thus, FWA is 10^{30} times more accurate than linear algebraic models.

Conclusion

FWA treats the electron as a stable field node, whose parameters and interactions arise from the structure of the field and the distinguishability depth L_u . This dynamic algebra yields:

- dramatically higher accuracy,
- variable electron properties,
- field-dependent interactions,
- and new experimental predictions.

FWA therefore surpasses linear algebraic models by many orders of magnitude.